Time Value of Money

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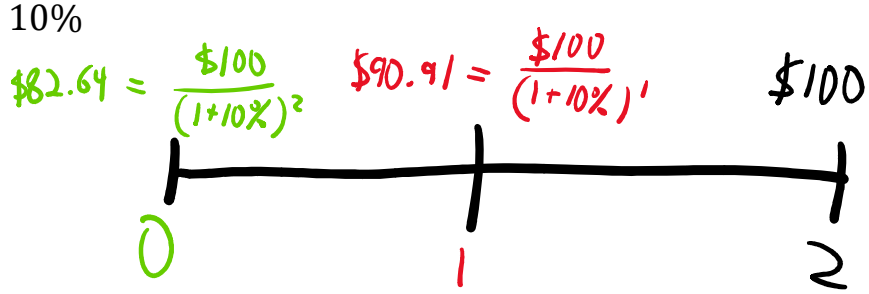
Present & Future Value Terminology

- Present Value: value of cash flows today
- Future Value: value of cash flows in the future
- Discount rate: relates the value of money across time periods and risky states

- $r = 10\%$



Tools

We will learn how to solve these problems:

- Analytically with formulas: I do not want you to view Excel as a black box
- Excel: I want you to be efficient

Tips

- Do not round intermediate calculations
- Draw timelines
  - Help you see where cash flows happen
  - Give you a sense of what info you have, and what you will be solving for
- Interest rates and periods need to be in sync
  - If you're talking about monthly cash flows, then interest rate and periods needs to be in monthly terms

TVM Formulas

- Present value of future cash flow

$$PV_0 = \frac{FV_{\mathrm{t}}}{(1+r)^t}$$

- Present value of annuity

$$PV_0 = PMT \times \underbrace{\left[\frac{1-\left(\frac{1}{1+r}\right)^n}{r}\right]}_{ADF}$$

- Future value of annuity

$$FV_n = PMT \times \underbrace{\left[\frac{(1+r)^n - 1}{r}\right]}_{ACF}$$

- Perpetuity

$$PV_0 = \frac{PMT_1}{r-g}$$

Interest Rate Formulas

- APR: Annual Percentage Rate
  - Stated, annualized rate
  - Not a true cost of borrowing; simple interest
  - Compounding frequency given with n periods per year
- EAR: Effective Annual Rate
  - True cost of borrowing on an annualized frequency
- IRR: Internal rate of return
  - r effective period rate

$$1 + APR = 1 + r \times n$$
$$1 + EAR = (1+r)^n$$

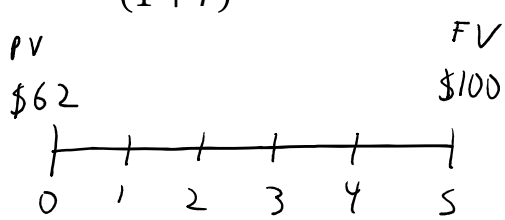
Excel TVM Functions

TVM Formulas for Excel:

- $PV = CF_0 + NPV(r, CF_t)$
- $PV = PV(r, n, PMT, FV)$
- $FV = FV(r, n, PMT, PV)$
- $PMT = PMT(r, n, PV, FV)$
- $r = IRR(CF)$
- $r = RATE(n, PMT, PV, FV)$

Present Value of Single Cash Flow

$$PV_0 = \frac{FV_\mathrm{t}}{(1+r)^t}$$



Example:

- Receive $100 in 5 years
- Discount rate 10%
- Present value today?

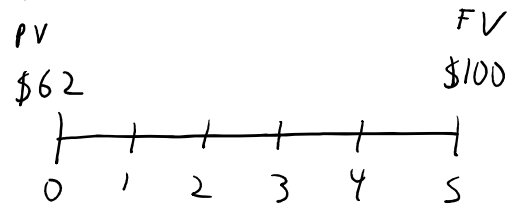
$$PV_0 = \frac{\$100}{(1+10\%)^5} = \$62.09$$

Excel

- $PV(r = 10\%, n = 5,, FV = 100) = 62.09$

Future Value of a Single Cash Flow

$$PV_0 = \frac{FV_\mathrm{t}}{(1+r)^t} \Leftrightarrow FV_t = PV_0 \times (1+r)^t$$



Example:

- Invest \$62.09 today
- Discount rate 10%
- Future value in year 5?

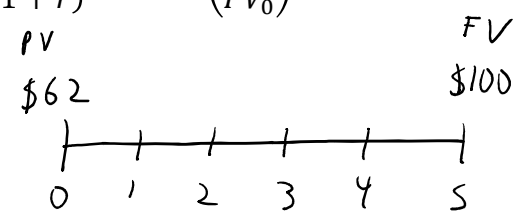
$$FV_5 = \$62.09 \times (1+10\%)^5 = \$100$$

Excel

- $FV(r = 10\%, n = 5,, PV = 62.09) = 100$

Internal Rate of Return (IRR) of a Single Cash Flow

$$PV_0 = \frac{FV_\mathrm{t}}{(1+r)^t} \Leftrightarrow r = \left(\frac{FV_t}{PV_0}\right)^{1/t} - 1$$



Example:

- Pay \$62.09 today
- Receive \$100 in 5 years
- What is the return?

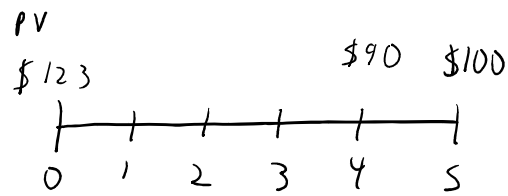
$$r = \left(\frac{\$100}{\$62.09}\right)^{1/5} - 1 = 10\%$$

Excel

- $RATE(n = 5,, PV = -62.09, FV = 100) = 10\%$

Present Value of Multiple Cash Flows

$$PV_0 = \sum_t \frac{FV_t}{(1+r)^t}$$



Example:

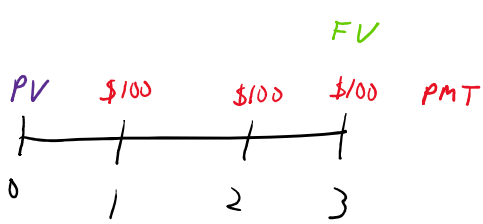
- Receive $90 and $100 in 4 and 5 years
- Discount rate 10%
- Present value today?

$$PV_0 = \frac{\$90}{(1+10\%)^4} + \frac{\$100}{(1+10\%)^5} = \$123.56$$

Annuity

- An annuity is an asset that pays a fixed cash flow for a fixed number of time periods
  - The price is the present value of the payments at time 0
  - The default timing assumption is that the first payment occurs at time 1

- For example: You will receive $100 per year for 3 years



Annuity – Formulas

$$PV_0 = PMT \times ADF(n, r)$$

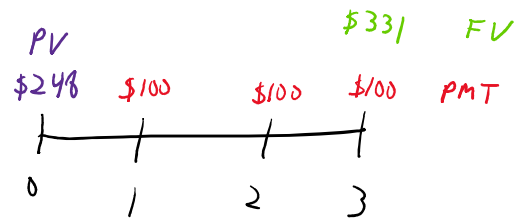
$$ADF(n, r) = \left[\frac{1 - \left(\frac{1}{1+r}\right)^n}{r}\right]$$

$$FV_n = PMT \times ACF(n, r)$$

$$ACF(n, r) = \left[\frac{(1+r)^n - 1}{r}\right]$$

Annuity Example

- Receive \$100 per year for three years
- Discount rate 10%



- Present value of annuity

$$\begin{aligned} PV_0 &= \$100 \times ADF(n = 3, r = 10\%) \\ &= PV(r = 10\%, n = 3, PMT = \$100) \\ &= \$248.69 \end{aligned}$$

- Future value of annuity

$$\begin{aligned} FV_3 &= \$100 \times ACF(n = 3, r = 10\%) \\ &= FV(r = 10\%, n = 3, PMT = \$100) \\ &= \$331 \end{aligned}$$

Annuity – Payments

- Pay \$248.69 for a 3-year annuity
- Discount rate 10%
- What are the payments?

$$PV = PMT \times ADF(n, r)$$

$$\$248.69 = PMT \times \left[\frac{1 - \left(\frac{1}{1 + 10\%}\right)^3}{10\%}\right]$$

$$\Rightarrow PMT = \$100$$

Excel

- $PMT(r = 10\%, n = 3, PV = -248.69) = 100$

Annuity – IRR

- Pay \$248.69 for a 3-year annuity
- Receive \$100 per year for three years
- What is the return?

$$PV = PMT \times ADF(n, r)$$

$$\$248.69 = \$100 \times \left[\frac{1 - \left(\frac{1}{1 + r}\right)^3}{r}\right]$$

$$\Rightarrow r = 10\%$$

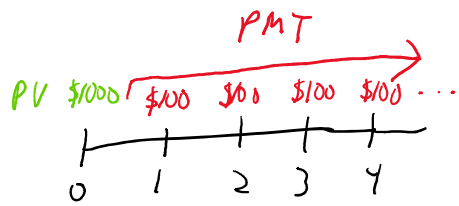
Excel

- $RATE(n = 3, PMT = 100, PV = -248.69) = 10\%$

Perpetuity

- A perpetuity is an asset that pays a fixed cash flow for an infinite number of time periods
  - The price is the present value of the payments at time 0
  - The default timing assumption is that the first payment occurs at time 1
  - Payments may grow at fixed growth rate g

$$PV_0 = \frac{PMT_1}{r - g}$$



Example:

- Receive $100 per year, forever
- Discount rate 10%
- Present value today?

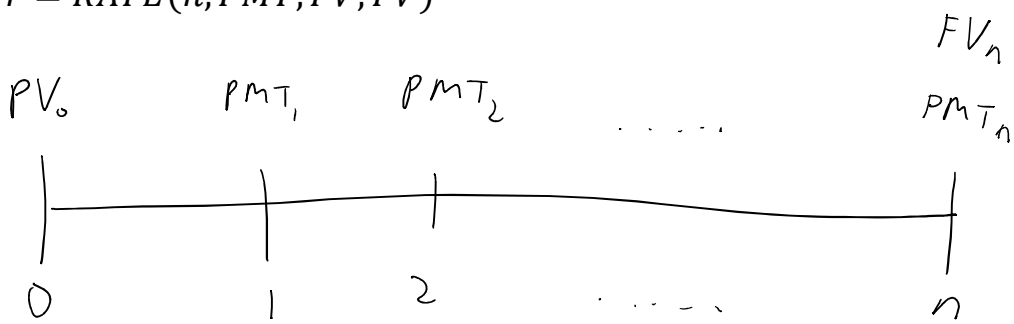
$$PV_0 = \frac{\$100}{10\%} = \$1000$$

Excel TVM Functions

- $PV = CF_0 + NPV(r, CF_t)$
- $PV = PV(r, n, PMT, FV)$
- $FV = FV(r, n, PMT, PV)$
- $PMT = PMT(r, n, PV, FV)$
- $r = IRR(CF)$
- $r = RATE(n, PMT, PV, FV)$

Standard TVM Functions

- $PV = PV(r, n, PMT, FV)$
- $FV = FV(r, n, PMT, PV)$
- $PMT = PMT(r, n, PV, FV)$
- $r = RATE(n, PMT, PV, FV)$



Timing type

Optional input *type* that controls timing of first payment, e.g.

- $PV = PV(r, n, PMT, FV, type)$
- Default: *type*=0 ⟹ First payment at time 1
- Override: *type*=1 ⟹ First payment at time 0
- Note PV and FV do NOT move

Generic TMV Functions

If we have an irregular stream of cash flows, we can use:

- $PV = CF_0 + NPV(r, CF_t)$
- $r = IRR(CF)$

Interest Rate Formulas

- APR: Annual Percentage Rate
  - Stated, annualized rate
  - Not a true cost of borrowing; simple interest
  - Compounding frequency given with n periods per year
- EAR: Effective Annual Rate
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- IRR: Internal rate of return
  - r effective period rate

$$1 + APR = 1 + r \times n$$
$$1 + EAR = (1 + r)^n$$

Example 1

Loan with 12% annual rate, compounded monthly

- $APR = 12\%$
- Compounded monthly: $n = 12$ periods per year
- Monthly effective rate: $r = 1\%$

$$1 + 12\% = 1 + r \times 12$$
$$\Rightarrow r = \frac{12\%}{12} = 1\%$$

- Effective Annual Rate: $EAR = 12.7\%$

$$1 + EAR = (1 + 1\%)^{12}$$
$$EAR = 12.7\%$$

Example 2

Your opportunity cost of capital is 6% per year, compounded quarterly. Calculate the following effective rates: month, quarter, annual.

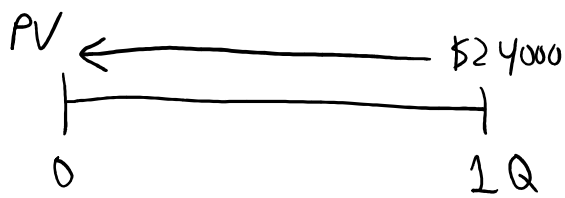
$$\text{quarterly rate} = \frac{6\%}{4} = 1.5\%$$
$$\text{monthly rate} = (1 + 1.5\%)^{\frac{1}{3}} - 1 = 0.4975\%$$
$$\text{annual rate} = (1 + 1.5\%)^{4} - 1 = 6.1364\%$$

Example 3

Your opportunity cost of funds is 6% per year, compounded quarterly. What is the value today of an internship that pays $24,000 in three months?

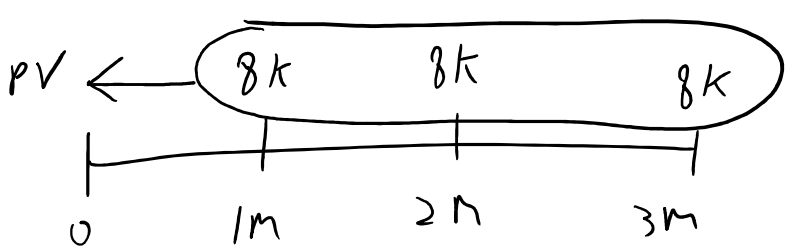
$$\text{quarterly rate} = \frac{6\%}{4} = 1.5\%$$

$$PV = \frac{\$24{,}000}{(1 + 1.5\%)^1} = \$23{,}645$$



Example 4

Your opportunity cost of funds is 6% per year, compounded quarterly. What is the value today of an internship that pays $8,000 per month for three months?



Example 4

Your opportunity cost of funds is 6% per year, compounded quarterly. What is the value today of an internship that pays $8,000 per month for three months?

$$\begin{aligned}\text{monthly rate} &= 0.4975\% \\ \text{PV} &= PMT \times ADF \\ &= \$8{,}000 \times \left[\frac{1 - \frac{1}{(1+0.4975\%)^3}}{0.4975\%}\right] \\ &= \$23{,}763\end{aligned}$$

IRR

The IRR can be found with the Excel functions

- $r = IRR(CF)$
- $r = RATE(n, PMT, PV, FV)$

IRR Example

You have been offered the following opportunity: Receive a tax credit of $10,000 today, and in return, you must make higher tax payments of $1,000 per year for 15 years. The first of the annual payments will come in one year's time. What is the internal rate of return on this investment?

$$PV = PMT \times ADF(n, r)$$

$$\$10{,}000 = \$1{,}000 \times \left[\frac{1 - \frac{1}{(1+r)^{15}}}{r}\right]$$

$$\Rightarrow r = 5.56\%$$

Excel

- $IRR(-10000{,}1000, \cdots, 1000\ ) = 5.56\%$
- $RATE(n = 15, PMT = 1000, PV = -10{,}000) = 5.56\%$